

MARK PRAINITO

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EDUCATION

Harvard College

Cambridge, MA

A.B. Candidate, Joint Concentration in Chemistry & Physics and Engineering Sciences (Electrical and Computer Engineering);
Secondary in Mathematics | Expected May 2029

Select Coursework: Laboratory Electronics, Probability, Microeconomics, Quantum Mechanics, Statistical Mechanics, Linear Algebra

TECHNICAL EXPERIENCE

Harvard University

Cambridge, MA

Course Assistant, Physics 123/223: Laboratory Electronics

August 2026 – Present

- Selected as the sole course assistant for a cross-listed undergraduate/graduate laboratory electronics course; support instruction, grading, and debugging in analog and digital circuit design.

Creative Nanobiophotonics Lab, Harvard University / Rice University

Cambridge, MA / Houston, TX

Researcher; PI: Professor Sangyeon (Fred) Cho

November 2025 – Present

- Design nanolaser systems for advanced microscopy and photonic computing; experimentally validate a light-controlled chemical reservoir exhibiting nonlinear dynamics and fading memory.
- Develop eight physics- and chemistry-informed reservoir-computing architectures and scalable Python, MATLAB, and COMSOL workflows for billions of simulations across time-series and image-classification benchmarks.
- Develop a hemoglobin-based computing platform for in situ biological monitoring; lead preparation of a first-author manuscript and contribute to an associated patent application.

Harvard Weitz Lab

Cambridge, MA

Research Assistant; PI: Professor David A. Weitz

November 2025 – Present

- Conduct microfluidic emulsion generation and characterization, supporting NMR-based measurements, experimental instrumentation, and quantitative data analysis.

Harvard College Students for the Exploration and Development of Space (SEDS)

Cambridge, MA

Bus-Mechanical Engineering Subteam Member

September 2025 – Present

- Designed and manufactured CubeSat mechanical hardware using CAD and structural/thermal analysis; coordinated electrical and software integration for an April 2026 SpaceX Falcon 9 launch.

Harvard Rocket Propulsion Group (HRPG)

Cambridge, MA

Engineer

September 2025 – Present

- Develop analytical and computational models supporting rocket-engine performance, propellant-flow analysis, and subsystem integration.

TECHNICAL SKILLS

Programming & HDL: Python, MATLAB, C++, Verilog

Engineering & Modeling: SolidWorks, Altium, COMSOL; analog/digital circuits, oscilloscopes, structural/thermal analysis

Laboratory: Nanofabrication, reactive-ion etching, SEM, NMR-based characterization, microfluidics

LEADERSHIP EXPERIENCE & HONORS

Harvard-MIT Mathematics Tournament

Cambridge, MA

Outreach Director

September 2025 – Present

- Lead Harvard-side outreach, partnerships, communications, and event logistics for programs serving 1,000+ participants, including collaborations with competitions at Stanford, Princeton, Yale, and Carnegie Mellon.

Empowered Innovations Inc.

Bellmore, NY

Founder & Co-President

January 2024 – Present

- Founded a nonprofit that designs and 3D-prints assistive devices for dozens of people with disabilities; secured support from two filament manufacturers.

Honors: Regeneron Science Talent Search Scholar (2025); Eagle Scout